Fall Detection Doppler Radar System Based on Time and Spatial Division

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***Abstract*—This work demonstrates a 2.45-GHz beam-switching Doppler radar system for real-time human fall detection. By utilizing Butler matrixes for beam switching, the system divides the spatial coverage to distinguish different body regions. Combined with time-sequence information, it enables the continuous tracking of human motion, providing more comprehensive data for fall detection and analysis. Experimental results validate the feasibility and reliability of the proposed system.**

Keywords—Doppler radar, beam switching, human motion, fall detection.

I. Introduction

Human motion analysis has a broad range of applications, such as medical diagnostics, healthcare, sports training, and rehabilitation. Body movement consists of a series of actions performed by interconnected parts at varying speeds. Consequently, observing the conditions of different body parts can obtain meaningful information about motion. Among these applications, senior healthcare has gained significant attention due to the challenges of an aging society. For the elderly, accidental falls can result in severe and often irreparable harm. One common approach for fall detection involves using high-speed video to capture frame-by-frame images [1]. Processing a substantial volume of images requires intensive computational resources to extract motion features. Moreover, it exists a privacy issue. Another widely adopted method is based on accelerometers, which necessitate attaching sensors to specific body parts [2].

Over the past decade, radar technology has been widely used to capture and track human motions [3]-[5]. This technology utilizes the Doppler effect, a relative displacement at different body parts will introduce frequency modulation to the transmitting signals. By extracting the frequency shift, information on the velocity and direction of an object can be obtained.

[4] proposed a 5.8 GHz FMCW radar system to detect fall occurrences and distinguish them by analyzing Doppler, range, and radar cross-section (RCS) data extracted from real-time ISAR imaging. In [5], The Doppler information from radar was used for the analysis of fall detection based on a deep learning-based approach. The radar technology has been proven to be an effective method for fall detection. However, the motion of the body is a sequence of actions performed by interconnected body parts with varying speeds. Therefore, segmenting body parts is essential for motion analysis. Most existing literatures utilize single-input single-output (SISO) radar systems, which may encounter a challenge during the signal processing. When dealing with more complex body movements, it requires significant effort to distinguish different actions of body parts.

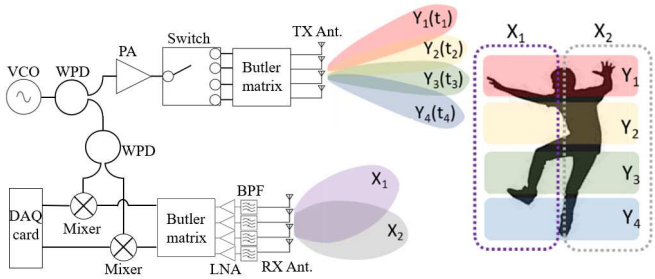

Fig. 1. The concept of beam spatial division for corresponding body parts.

This work proposes a beam-switching Doppler radar system for real-time human fall detection. The system employs Butler matrixes to enable beam scanning and enhance spatial diversity along with time sequency. As illustrated in Fig. 1, the transmitter sequentially switches between the four beams (Y_1–Y_4) from top to bottom. At the receiver, dual-beams simultaneously capture reflected waves from both sides (X_1, X_2). This configuration divides the spatial beams into eight regions, corresponding to the bilateral sections of the human body: the head, torso, thighs, and calves. By analyzing the Doppler effect from individual body parts, sequential motion patterns can be extracted for fall detection.

II. System Design

Figure 1 also shows the block diagram of the proposed radar system. A voltage-controlled oscillator (VCO) generates a 2.45-GHz continuous wave (CW) signal, which is amplified by a power amplifier (PA). The amplified signal is sequentially fed through input ports to a 4×4 Butler matrix, enabling the generation of four beams that scan in different directions, each targeting specific regions of the body. Reflected signals, originating from these body regions, are captured by a dual-beam antenna array. The received signals pass through band-pass filters (BPF) to suppress environmental noise and are subsequently amplified by low-noise amplifiers (LNA). The amplified signals are then down-converted by the mixers to produce I/Q signals. Finally, these signals are processed via a data acquisition card (DAQ) for post-signal analysis.

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TABLE I. PERFORMANCE ON EACH SUB-CIRCUIT

Component	Measured Parameter @ 2.45 GHz
4×4 Butler Matrix	Δθ = (-45°±3°; 135°±12°; -135°±9°; -45°±8°), IL = -7.6±1.6 dB;
Antenna	Bow-tie Ant. Gain=6.5 dBi, RL > 10 dB,
VCO (CVCO55BE-24002670)	P_{out} =5.5 dBm, Tuning Range = 2.4-2.65 GHz, Phase Noise = -100.35 dBc/Hz @ 1 MHz
Mixer (LT5575)	Gain=1.5 dB, Input P_{1dB}=11.5 dBm, Port-Port Isolation > 50 dB, RL > 11 dB
Switch (QPC6044)	IL = 1.2 dB, RL > 20 dB, Isolation > 50 dB Switch Time = 150 ns
Wilkinson Power Divider	IL = 3.2 dB, RL > 20 dB, Isolation > 30 dB
Power Amplifier (HMC414MS8GE)	Gain = 18.8 dB, RL = 33 dB/6 dB (I/O), OP_{1dB} = 27 dBm
Low-Noise Amplifier (HMC667)	Gain=19.3 dB, RL > 10 dB, NF = 0.8 dB
Bandpass Filter	IL=1.7 dB, RL > 15 dB

All the sub-circuits are fabricated using Rogers RO4003 printed circuit boards (ε_r=3.38, tanδ=0.0025). A summary of each sub-circuit's performance is listed in Table I.

The most important feature of the entire system is generating the desired radiation patterns. A bow-tie antenna was selected as the array element due to its 3-dB beamwidth of approximately 180° along the longitudinal plane. This wide beamwidth provides extensive horizontal coverage at the transmitter, ensuring that targets remain within the radar's field of view (FOV) even during sideways falls. On the receiver side, such a radiation pattern can span the full height of the target, from head to toe, achieving the required coverage. The measured gain of single bow-tie antenna is 6.5 dBi.

To enable vertical scanning from the transmitter, four antennas are vertically arranged with λ/2-spacing, achieving an array gain of up to 12 dBi. At the receiver, the antennas are arranged horizontally, providing an array gain of approximately 7.6 dBi. The signals from dual directions are simultaneously received through two output ports.

The measured array patterns are shown in Fig. 2. The transmitting array beams point at -44°, -14°, 12°, and 44°, corresponding to Y_1–Y_4 as illustrated in Fig.1. The reception pattern beams are directed at -14° and -16°, corresponding to X_1–X_2. The switch sequentially cycles through beams Y_1–Y_4, with each beam illuminating for 0.025 seconds, completing a full scanning cycle in 0.1 seconds. This brief duration, relative to human movement, allows for the capture of real-time dynamic motion. With a sampling rate of 100 kHz, 2500 data readings can be collected for each beam. After down-mixing, the received signals are processed to extract Doppler shifts. Further processing steps, including DC offset removal, averaging, and short-time Fourier transform (STFT), are applied to the raw data. The resulting time-frequency spectrogram provides insights into the motion dynamics over time and the corresponding body parts.

III. MEASUREMENT

Figure 3 illustrates the measurement setup, where the receiving antenna is positioned above the transmitting array. The distance between the radar and the target determines the illuminated area, allowing for adjustments to optimize signal reception. In this experiment, the distance was set to 1.25 m,

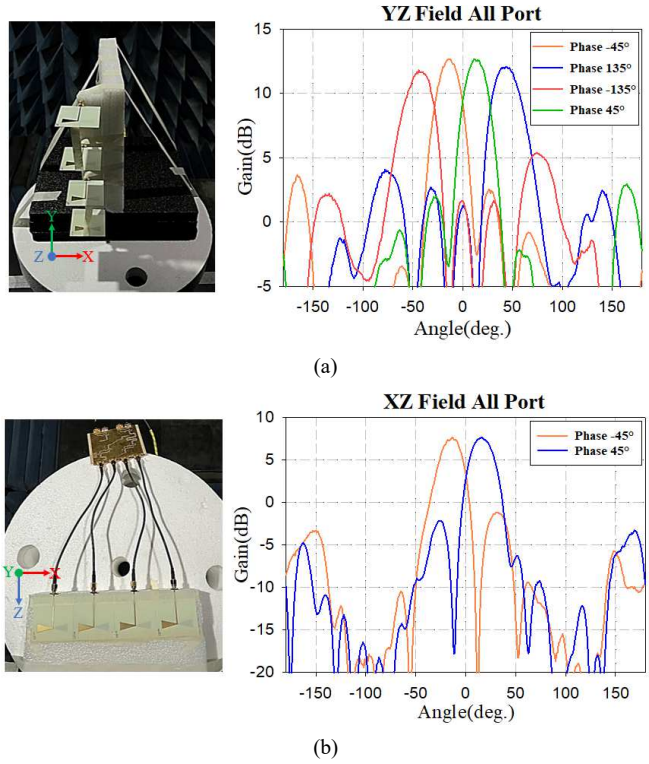

Fig. 2. Measured array patterns (a) Switched-beam transmitting array, (b) Dual-beam receiving array.

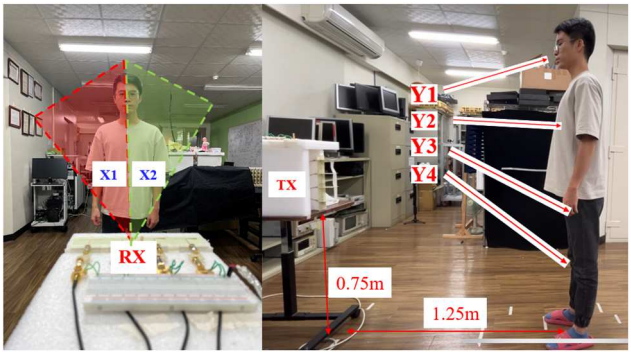

Fig. 3. The experiment setup.

ensuring that the radar's transmitted beam fully covers the test subject, who is approximately 170 cm tall.

The experiments were conducted for various simulated fall scenarios. Figure 4 shows the spectrogram of a sideways fall to the left. In this case, the subject falls toward the left side (X_2), gradually moving out of the coverage area of the right side (X_1). As a result, the signal strength in X_1 weakens significantly, while X_2 becomes the focus of the analysis. During time period t_1-t_2, the subject's body movement is not obvious, and the Doppler shift is minimal. Between t_2-t_3, as the body starts to fall and moves away from the radar, a negative Doppler shift occurs. During this period, the signal strength in Y_3 becomes noticeably stronger than in Y_2, mainly because the body moves downward and the falling speed increases. When the body reaches the ground, as the period of t_3- t_4, the legs arching forward due to the reflex action, generating positive Doppler shift in Y_4.

Figure 5 shows another scenario where the subject falls backward. In this case, signals are observed from both sides (X_1 and X_2) with similar signal intensity. Falling backward

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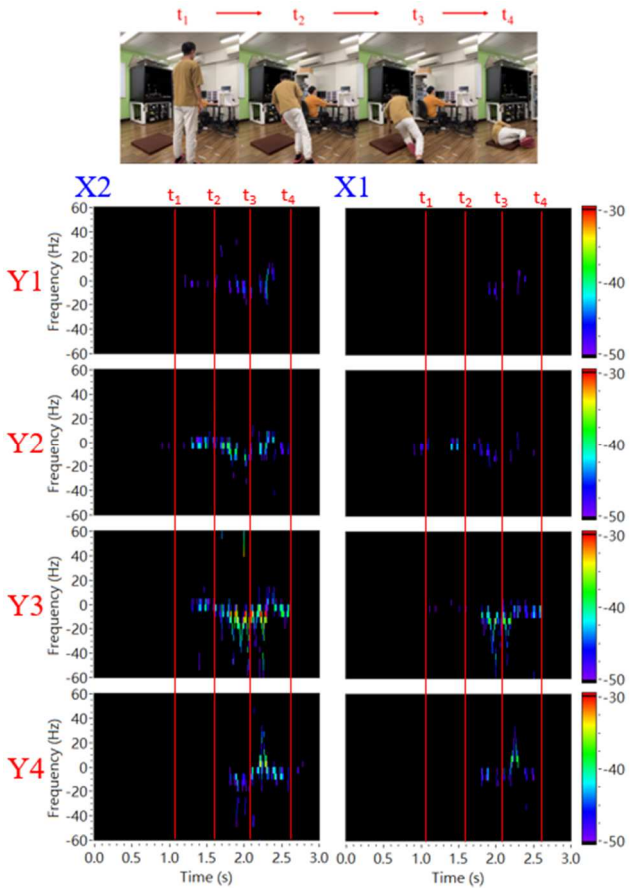

Fig. 4. Measurement results of fall detection (to the left).

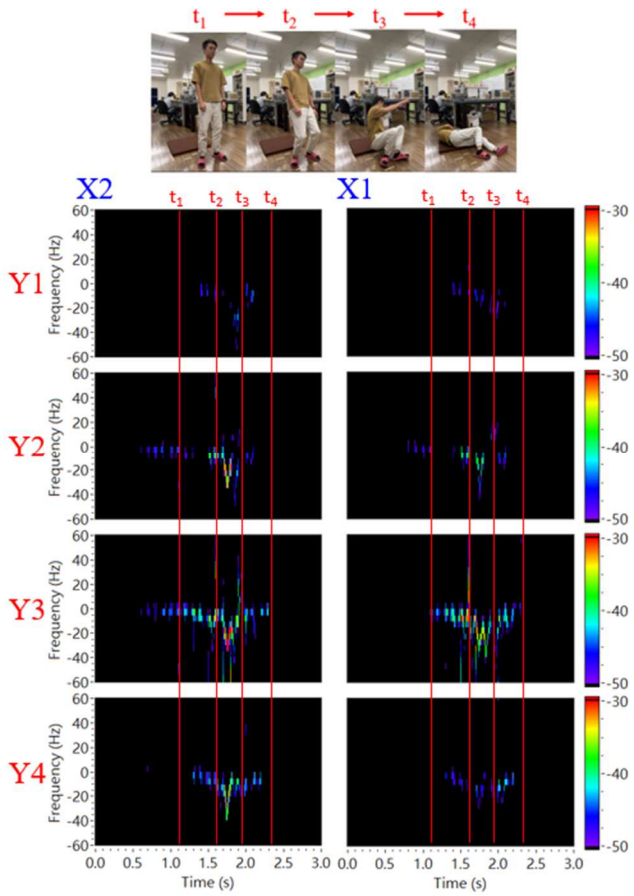

Fig. 5. Measurement results of fall detection (backward).

causes the entire body to move away from the radar, resulting in negative Doppler shifts across all windows. Especially during t_2-t_3, as the body rapidly falls, the Doppler shift reaches its maximum value.

IV. Conclusion

In an aging society, effective fall detection is crucial for enabling timely alerts and assistance. This paper introduces a 2.45-GHz beam-switching Doppler radar system capable of real-time human fall detection. By leveraging both spatial and temporal information, the system is able to capture continuous human motions. Currently, motion analysis is based solely on Doppler shifts. However, with the integration of AI-based training in the future, the system has great potential to achieve more accurate motion recognition and classification.

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